

Topic 1: States of matter - Answers

IGCSE Chemistry 0620 Paper 4 topical answer booklet.

Subtopic: Particle model and changes of state

Questions in this answer PDF: 6

Use this with the matching topical question PDFs. The answer points are organised by the same source labels.

Short explanation notes are added for harder calculation, electrolysis, equilibrium, rate, salts, organic, metals and practical questions.

Periodic table reference pages are not included to save printing paper.

2020 May/June Paper 42 Question 1

Main topic: Topic 1: States of matter | Marks: 11 | Answer source: 0620_s20_ms_42.pdf

Extra topic(s): Topic 2: Atoms, elements and compounds, Topic 8: The Periodic Table, Topic 9: Metals, Topic 4: Electrochemistry, Topic 7: Acids, bases and salts

Answer points

1(a)(i)

- condensation

1(a)(ii)

- sublimation

1(a)(iii)

- fractional distillation

1(a)(iv)

- electrolysis

1(a)(v)

- chromatography

1(b)(i)

- N

1(b)(ii)

- O

1(b)(iii)

- Be

1(b)(iv)

- N

1(b)(v)

- Ne

1(b)(vi)

- Li

Easy explanation / reminder

- *Electrolysis tip: cations go to the cathode and gain electrons. Anions go to the anode and lose electrons.*

- *Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.*

2020 May/June Paper 43 Question 5

Main topic: Topic 1: States of matter | Marks: 6 | Answer source: 0620_s20_ms_43.pdf

Extra topic(s): Topic 7: Acids, bases and salts

Answer points

5(a)

- particle
- separation
- particle
- arrangement
- type of
- motion
- solid
- touching
- liquid
- random

- gas
- random

5(b)

- condensing

5(c)

- evaporation
- diffusion

Easy explanation / reminder

- *Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.*

2021 Feb/March Paper 42 Question 2

Main topic: Topic 1: States of matter | Marks: 10 | Answer source: 0620_m21_ms_42.pdf

Extra topic(s): Topic 3: Stoichiometry, Topic 2: Atoms, elements and compounds

Answer points

2(a)

- chlorine

2(b)

- fluorine

2(c)(i)

- random motion of molecules / particles

2(c)(ii)

- hydrogen
- lowest (relative) molecular mass

2(d)(i)

2(d)(ii)

- argon / Ar

2(d)(iii)

- carbon dioxide

2(d)(iv)

- fractional (1)
- distillation (1)

Easy explanation / reminder

- *Practical tip: name the reagent/apparatus and give the positive observation. Observation marks need colour changes or precipitates clearly stated.*

2021 Oct/Nov Paper 42 Question 1

Main topic: Topic 1: States of matter | Marks: 11 | Answer source: 0620_w21_ms_42.pdf

Answer points

1(a)

- 1 mark for each correct row
- State
- touching
- random
- movement
- regularly
- arranged
- Gas
- ✓
- Liquid
- ✓
- ✓
- Solid
- ✓
- ✓

1(b)(i)

- boiling happens at a specific temperature (1)
- boiling has bubbles (1)

1(b)(ii)

- condensation (1)
- sublimation (1)

1(c)

- one horizontal line level with Y (1)
- two separate decreases before and after horizontal line (1)

1(d)(i)

- dissolving

1(d)(ii)

- precipitation

Easy explanation / reminder

- *Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.*
- *Metals tip: link the method to the reactivity series. Very reactive metals need electrolysis; less reactive metal oxides can often be reduced by carbon/CO.*

2022 Oct/Nov Paper 43 Question 2

Main topic: Topic 1: States of matter | Marks: 10 | Answer source: 0620_w22_ms_43.pdf

Extra topic(s): Topic 2: Atoms, elements and compounds, Topic 3: Stoichiometry

Answer points

2(a)(i)

- E

2(a)(ii)

- H

2(a)(iii)

- G

2(b)

- D (1)
- conducts electricity when solid (1)

2(c)

- F (1)
- high melting point (1)
- non-conductor of electricity when solid and liquid (1)

2(d)

- I (1)
- conducts electricity when liquid but not when solid / ONLY conducts when liquid (1)

2025 Oct/Nov Paper 42 Question 1

Main topic: Topic 1: States of matter | Marks: 13 | Answer source: 0620_w25_ms_42.pdf

Extra topic(s): Topic 10: Chemistry of the environment

Answer points

1(a)

- particle separation
- particle
- arrangement
- particle motion
- solid
- uniform
- vibrate
- liquid
- touching
- random
- random
- gas
- apart

1(b)

- melting
- condensing / condensation
- dissolving

1(c)(i)

- motion / movement of particles

1(c)(ii)

- relative molecular mass

1(d)

- photochemical smog
- respiratory problems

1(e)(i)

- carbon monoxide

1(e)(ii)

- nitrogen
- carbon dioxide

Easy explanation / reminder

- *Acids/salts tip: for soluble salts use excess insoluble base/carbonate, filter, then crystallise. For insoluble salts use precipitation.*
- *Organic tip: identify the functional group first. Combustion of hydrocarbons makes CO₂ and H₂O when oxygen is in excess.*